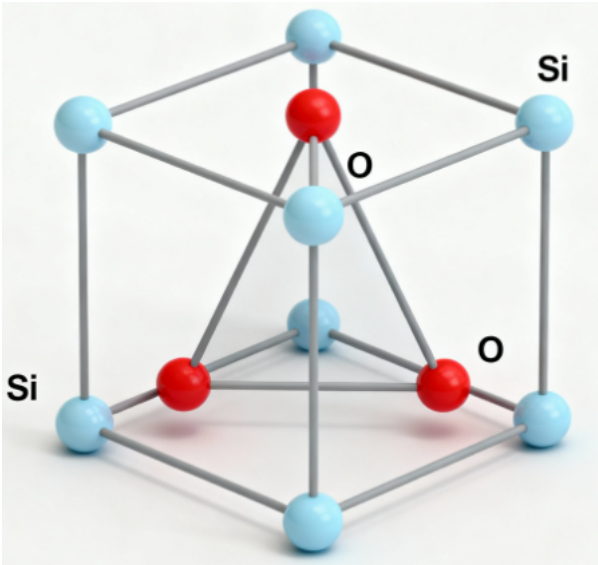
Purple amethyst

 SiO2 crystal lattice	**Characteristics:** Purple amethyst is a variety of quartz (SiO_2) with colors ranging from pale lilac to deep violet.These colors are caused by trace amounts of iron and natural irradiation. It commonly forms prismatic hexagonal crystals and fills geodes in igneous and metamorphic rocks. They are usually transparent to translucent with a vitreous luster and a Mohs hardness of 7. **Natural occurrences:** Purple amethyst occurs in hydrothermal veins, cavities, and geodes within igneous rocks like granite and basalt. It also occurs in metamorphic rocks under low to moderate temperatures. It forms where silica-rich fluids deposit quartz crystals, often associated with minerals like calcite and other quartz varieties. Major deposits are found in Brazil, Uruguay, India, Madagascar, and Zambia**.** **Fun facts:** Carvers and artisans make decorative items, sculptures, and home décor from amethyst.